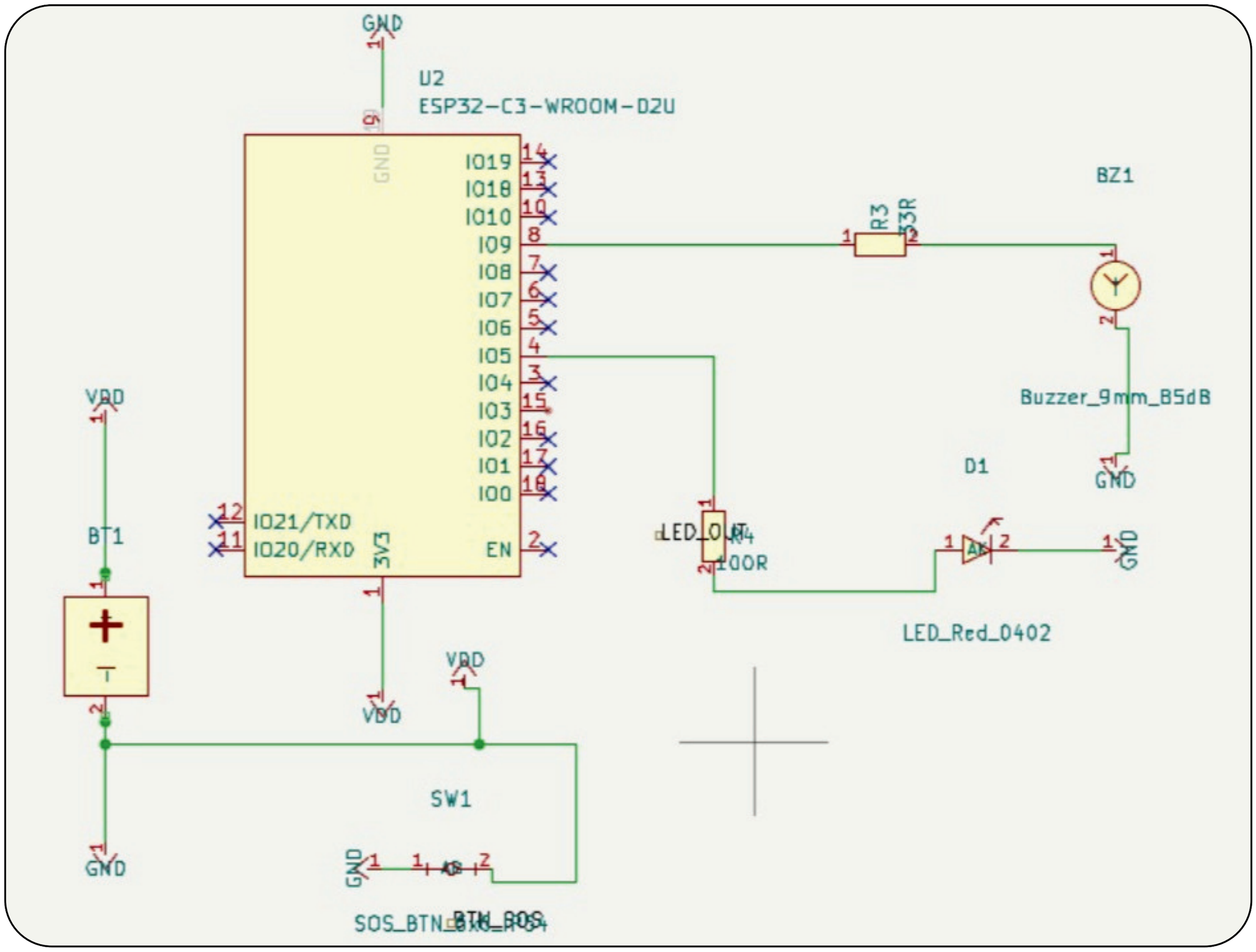
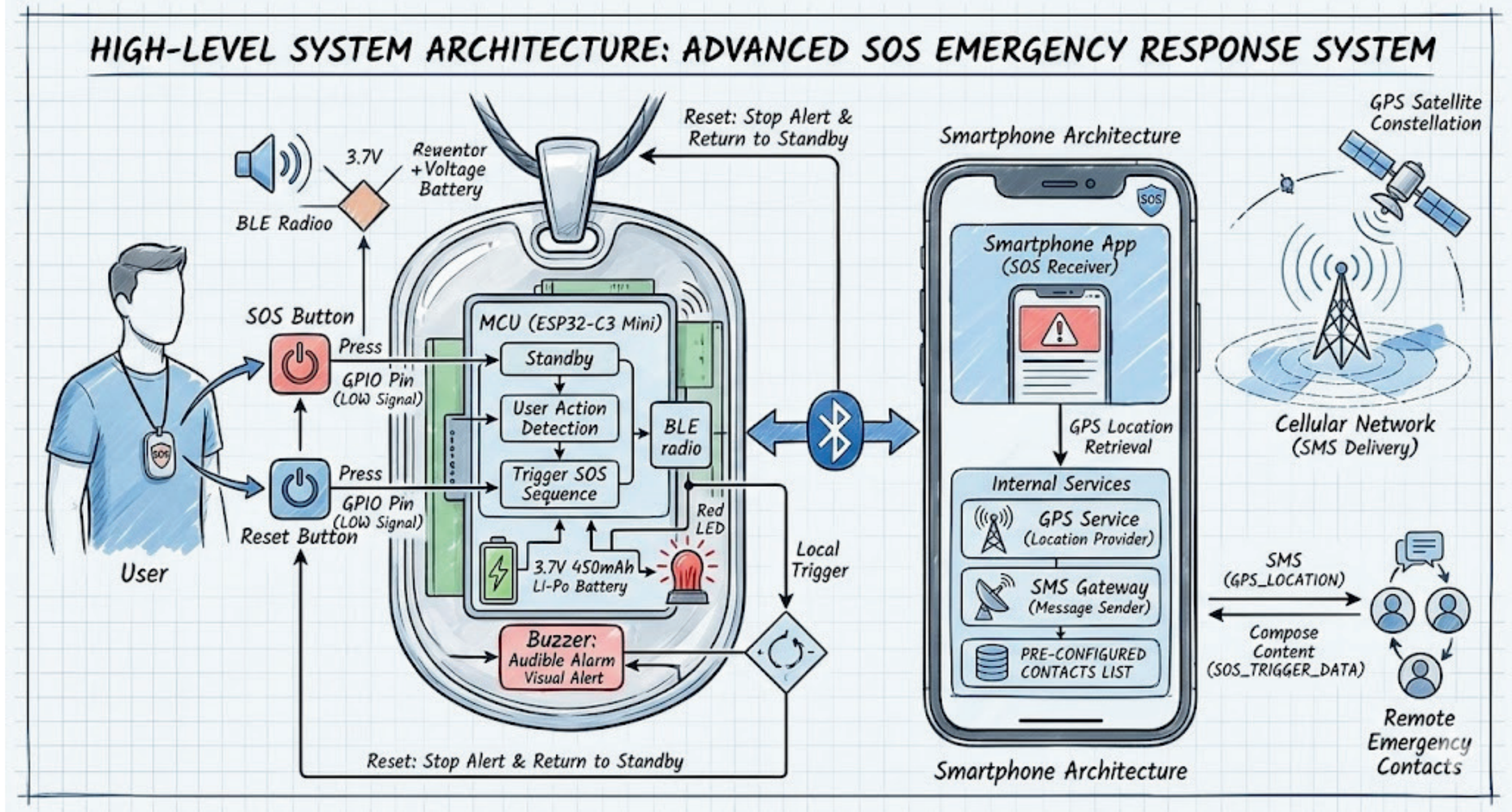
Circuit Diagram

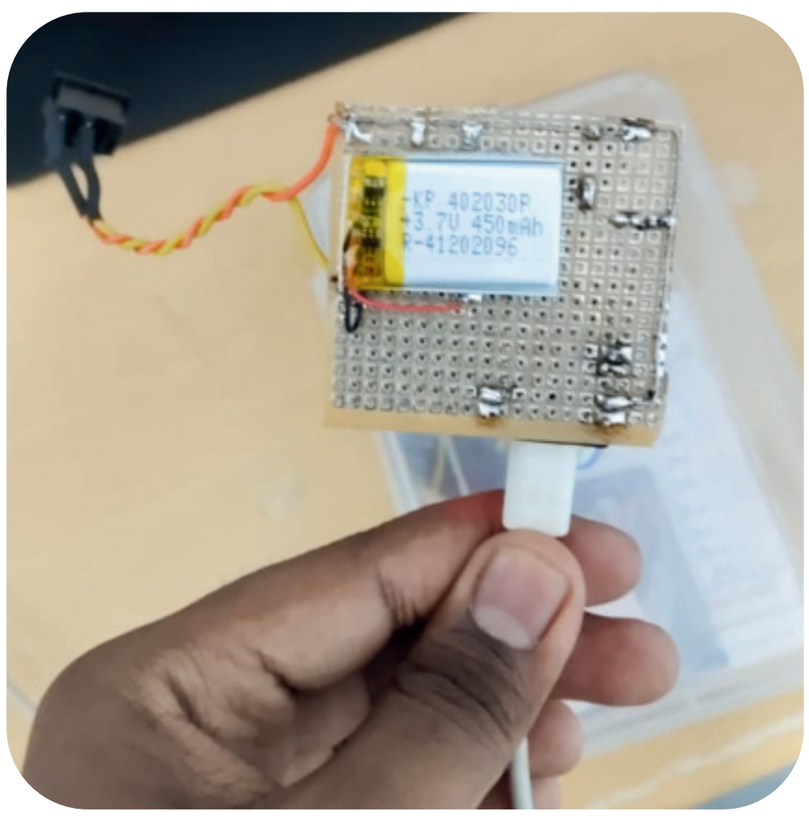


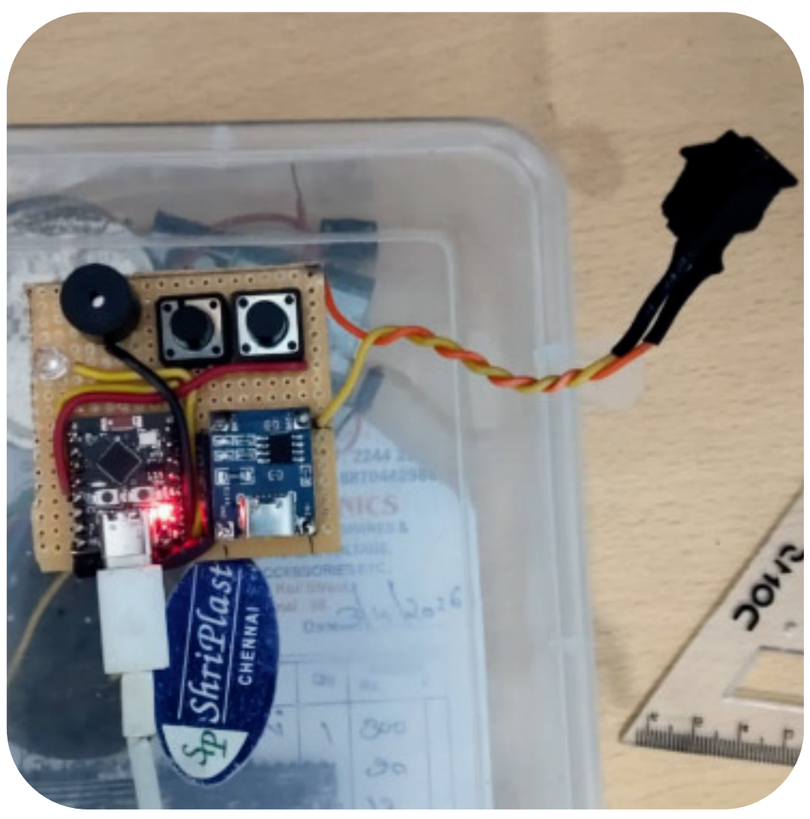
Core Components:

- **Processor :** **ESP32-C3 Mini** manages all logic, utilizing built-in **BLE** to communicate with the smartphone.
- **Alert System :** A synchronized **Active Buzzer** and **Red LED** provide immediate local audible and visual warnings.
- **Power Management :** A **1500mAh Li-Po battery** paired with a **TP4056 charging board** ensures safe, long-lasting portable power.
- **Interface :** Dual **Tactile Buttons** (SOS and Reset) connected to GPIO pins to trigger or cancel the emergency sequence.

Prototype Architecture







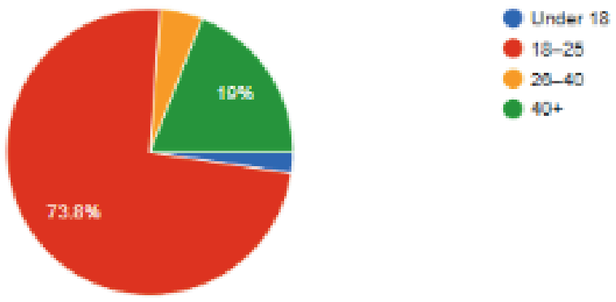
Working of the SOS Keychain Prototype:

1. **Activation:** User presses the **SOS button**, prompting the **ESP32-C3 Mini** to trigger the emergency sequence.
2. **Dual Alert:** The device immediately sounds a **loud buzzer** and flashes a **red LED** while simultaneously sending a **BLE signal** to the smartphone to dispatch a **GPS-linked SMS.**
3. **Cancellation:** The alerts remain active until the **Reset button** is pressed, returning the system to standby.

Customer Survey

What is your age group?

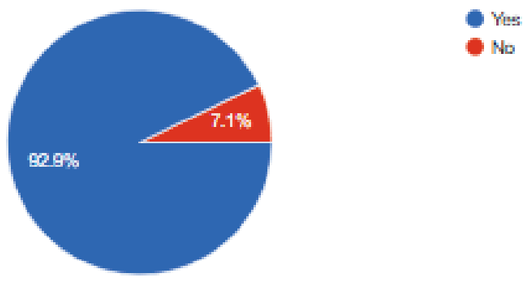
42 responses



Would you feel safer carrying a keychain-sized SOS device daily?

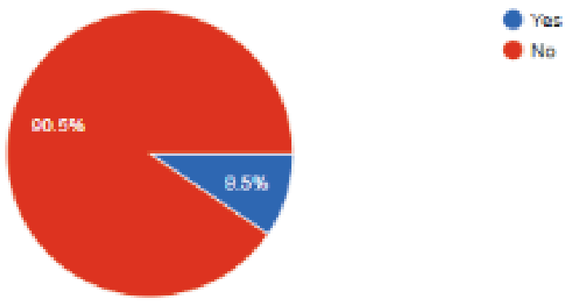
Copy chart

42 responses



Do you currently carry any personal safety device with you (e.g., pepper spray, whistle, small alarm)?

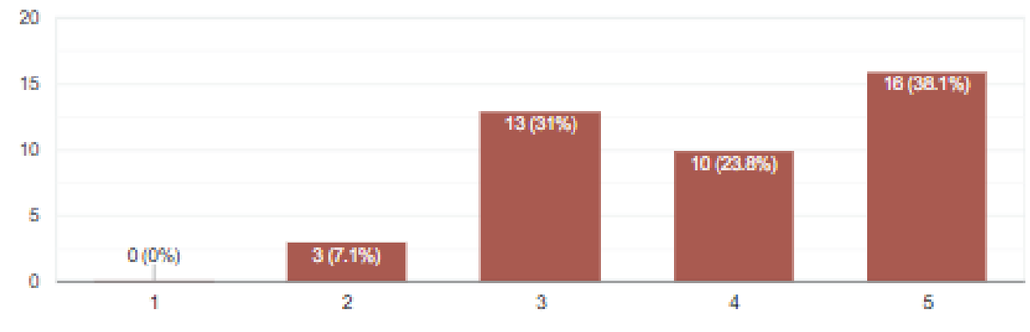
42 responses



Is the ability to send your live location to an emergency contact important to you?

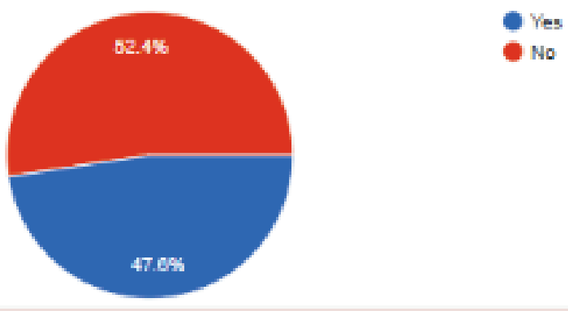
Copy chart

42 responses



Have you ever been in a situation where you wished you had an emergency alert device?

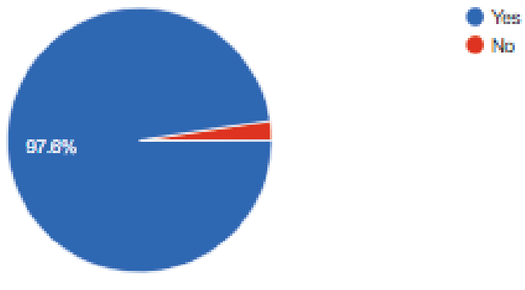
42 responses



Would you find it useful if the device automatically sent an SMS alert without needing a phone connection (e.g., uses its own cellular chip)?

Copy chart

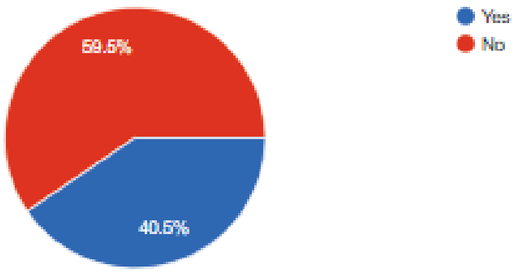
42 responses



Do you think a loud alarm alone is enough to deter an attacker or get attention in an emergency?

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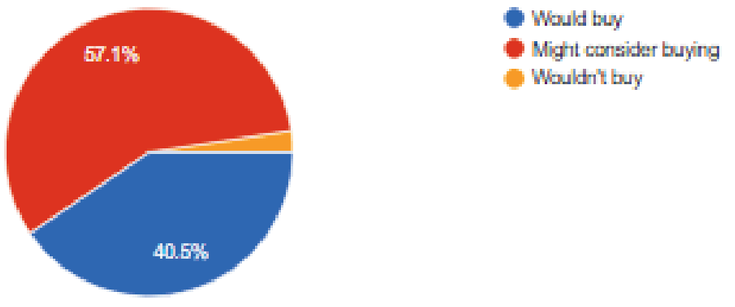
42 responses



How likely are you to buy this product if it were available and reasonably priced?

Copy chart

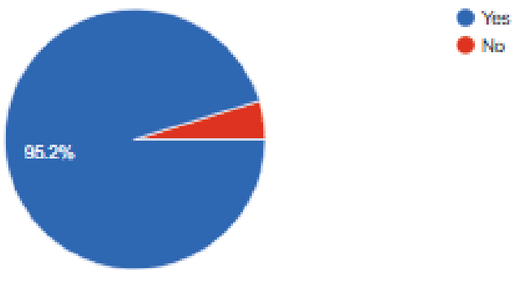
42 responses



Would knowing that professional help (e.g., security services) is automatically alerted give you more confidence in unsafe situations?

Copy chart

42 responses



Survey Result

The Personal Safety Device Interest Survey highlights a strong market interest among young adults for advanced personal safety solutions. Here's a concise analysis:

Key Takeaways

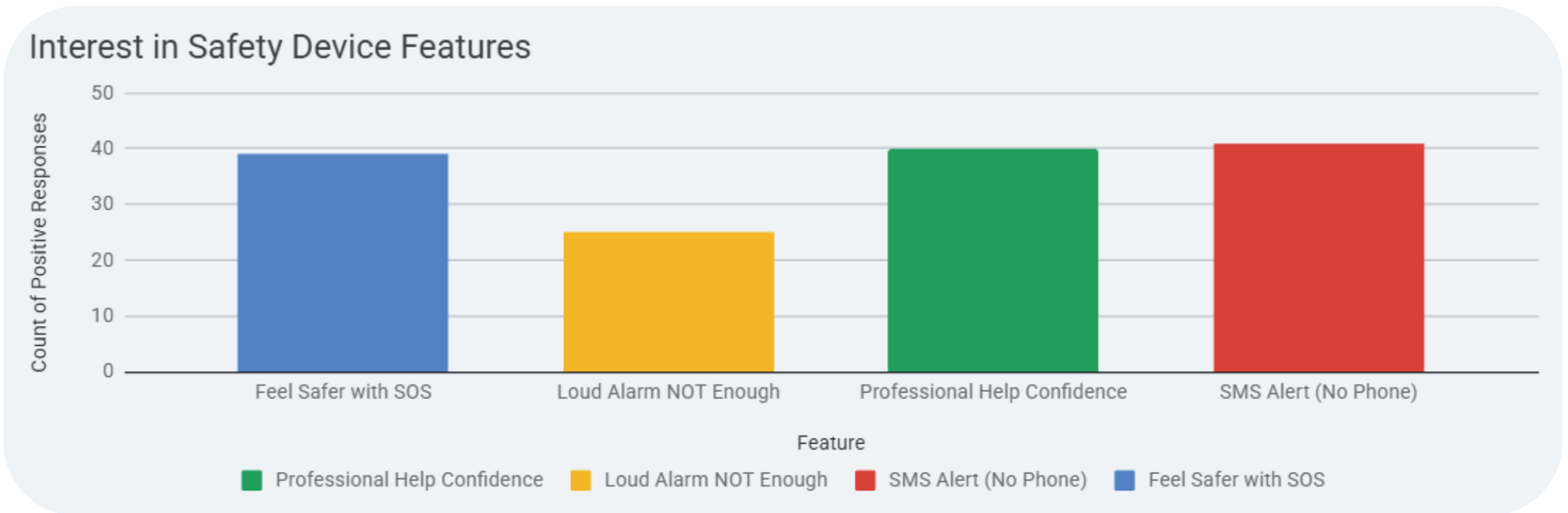
- **Strong Market Potential:**
  - 97.6% of respondents show a likelihood to purchase, with 17 committed to buying and 24 considering it.
  - Significant opportunity exists, with effective pricing and feature education potentially converting the *"Might consider buying"* segment.
- **Critical Feature Preferences:**
  - **Independent Connectivity:** 97.6% value SMS alerts without phone reliance.
  - **Professional Monitoring:** 95.2% desire automatic professional help alerts.
  - **Live Location:** High importance placed on live location sharing, rated 3.93 out of 5.
- **Demographic Focus:**
  - 73.8% of respondents are aged 18–25, indicating strong appeal among young adults.

Survey Insights

- **Demographics and Purchase Intent:**
  - The 18–25 age group is the primary demographic, suggesting targeted marketing strategies.
  - 67.1% are inclined to purchase if the device is reasonably priced.
- **Feature Interest:**
  - Independent SMS alerts and professional help are most desired.
  - Simple alarms deemed insufficient by the majority.

Summary

- **Safety Gap:** Young adults generally lack personal protection, despite previous vulnerability.
- **Essential Features:** Emphasis on automatic location tracking and emergency alerts.
- **Market Demand:** High interest in portable safety keychains, with a majority willing to purchase.



Product

SOS Keychain Device

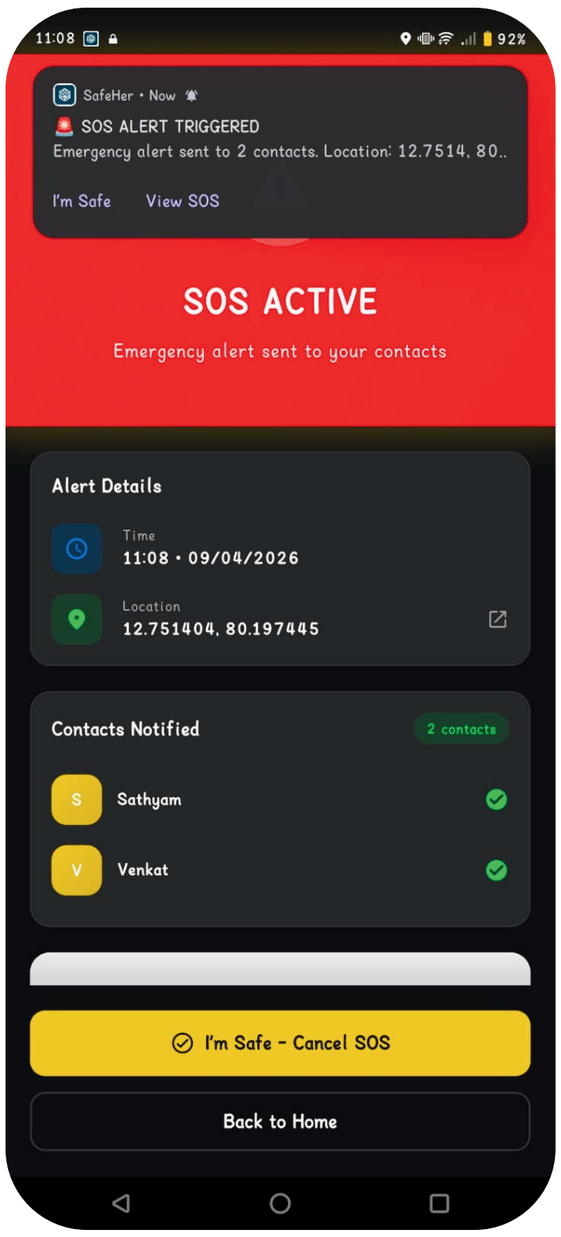


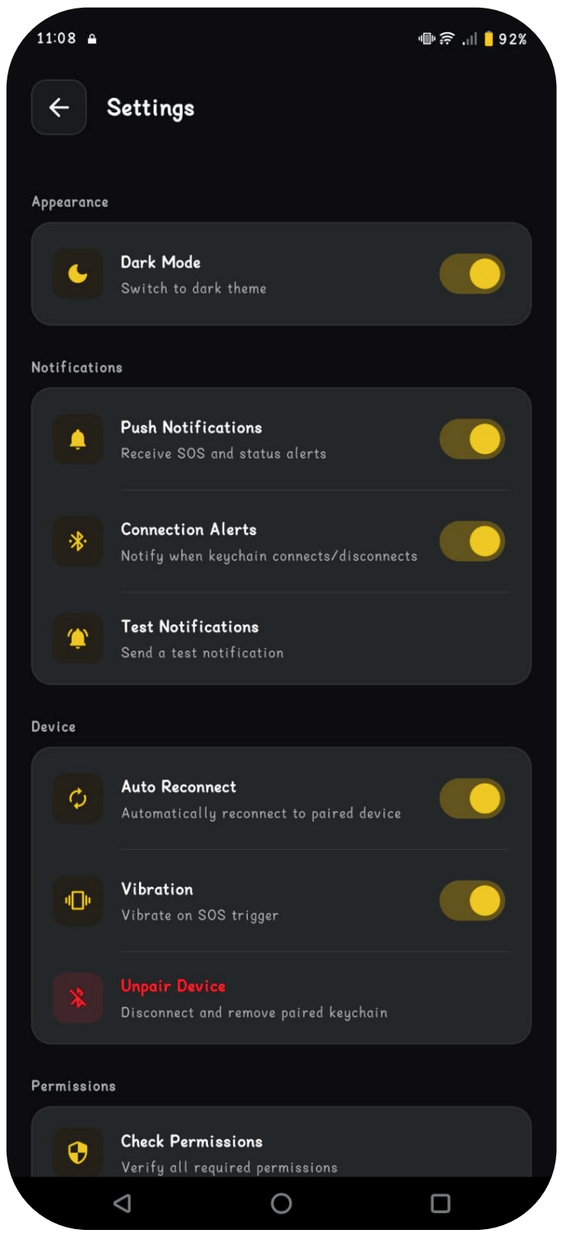


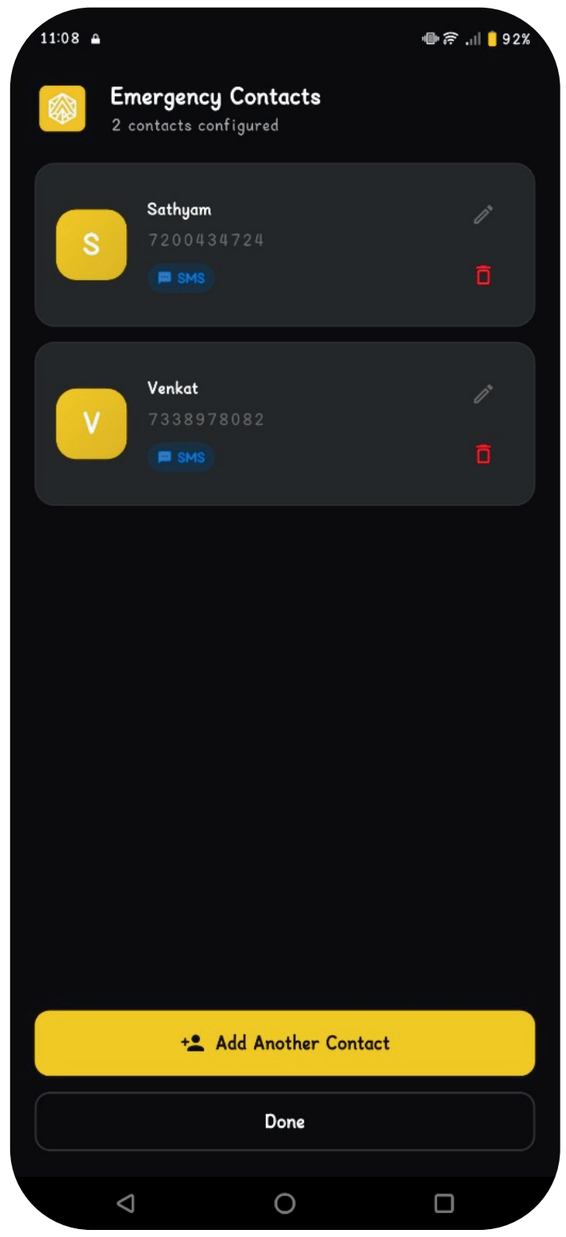


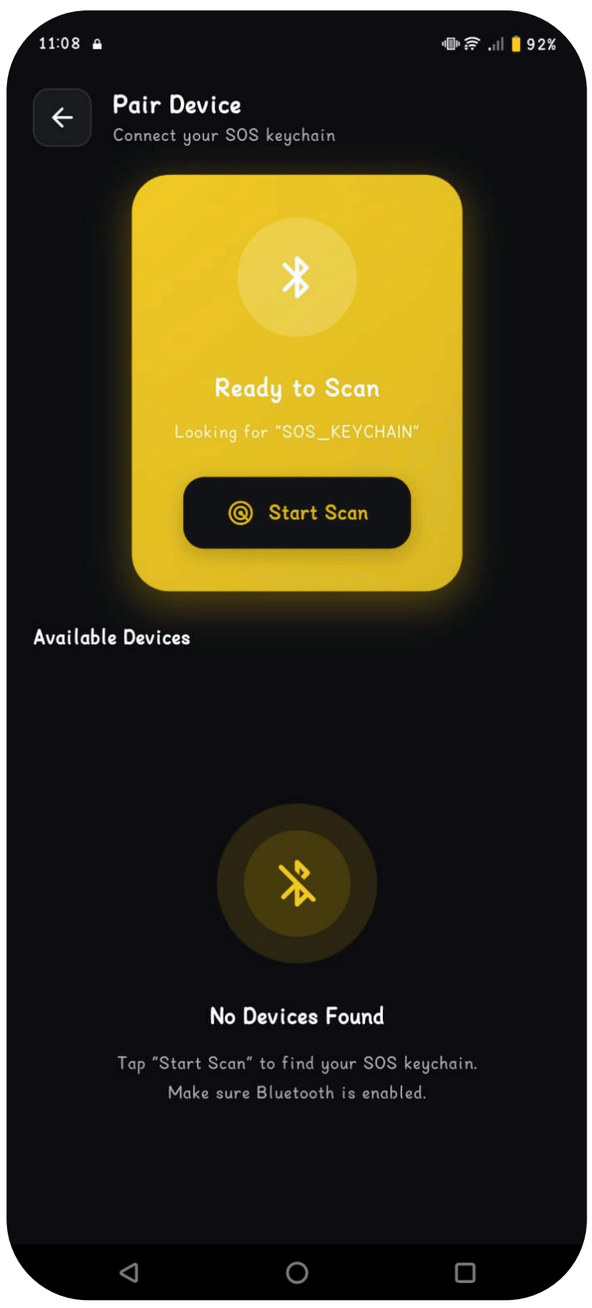


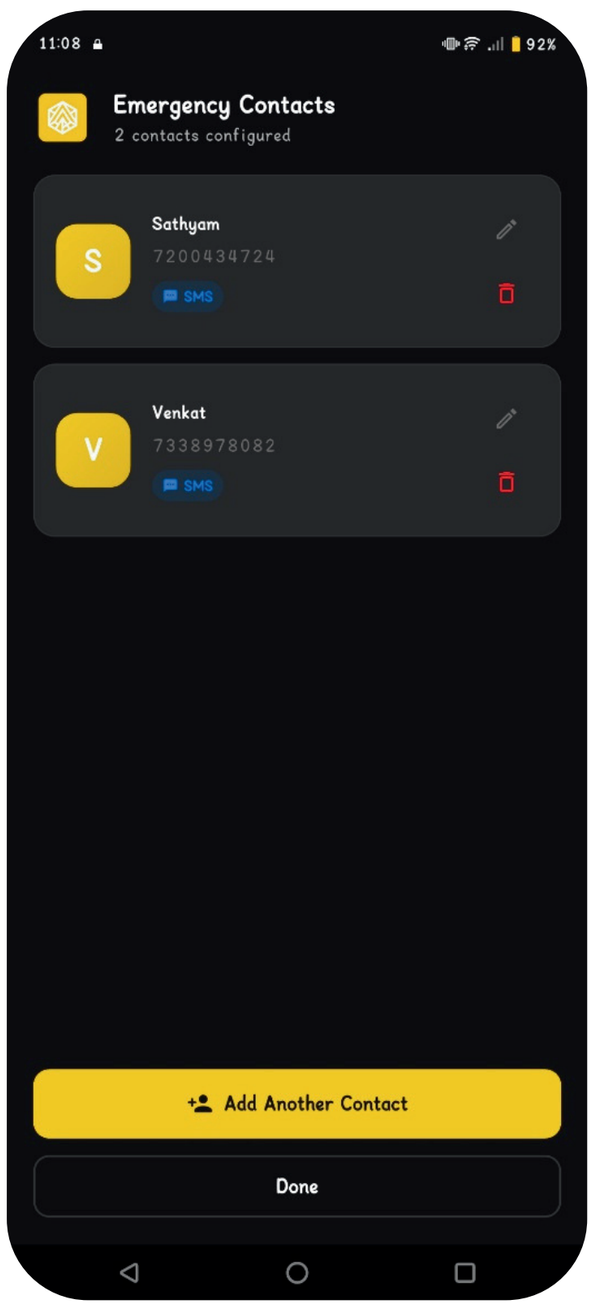
APP INTERFACE

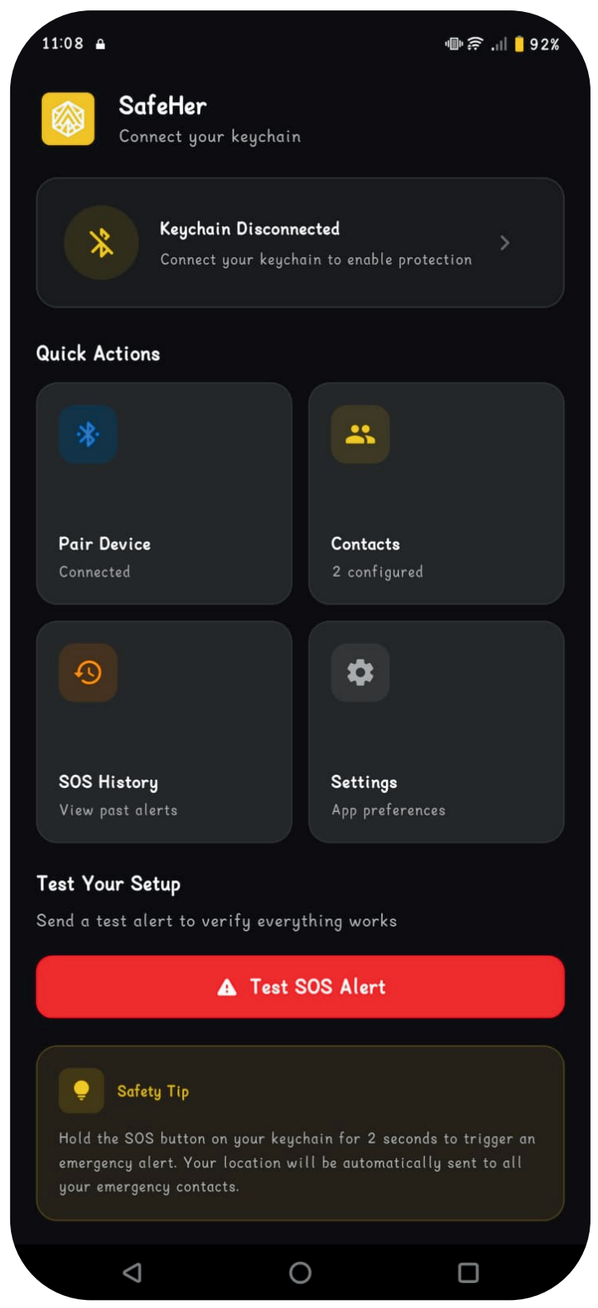












About Product

SafeHer : Integrated Safety System

The Keychain Device

- **Instant Trigger:** Portable 3D-printed hardware with a dedicated button for immediate emergency activation.
- **Local Alerts:** Features a loud built-in buzzer and a red LED to attract nearby help instantly.
- **Long-Lasting Power:** Driven by a rechargeable Li-Po battery designed for consistent standby protection.

The Companion App

- **Automated Response:** Syncs via Bluetooth to instantly send GPS location and SMS alerts to pre-set contacts.
- **Real-Time Monitoring:** Displays a live emergency dashboard with coordinates and connection status.
- **Smart Management:** Easy-to-use interface for contact setup, device pairing, and system testing.

